GEOGRAPHY

PHYSICAL GEOGRAPHY

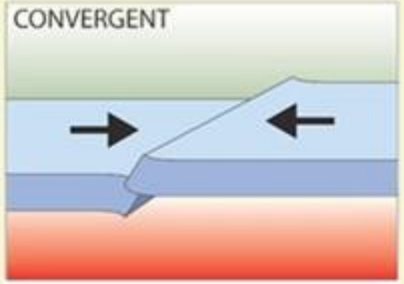
MODULE - 16

- SUBDUCTION ZONES
- RING OF FIRE
- EARTHQUAKES AND TSUNAMIS
- VOLCANOES

PREVIOUS LECTURE OF MODULE NUMBER-14

PLATE TECTONICS

2 DESTRUCTIVE / CONVERGENT BOUNDARY



1. This is a compressional or destructive boundary. Here the plates move towards each other.
2. They usually involve a continental plate and an oceanic plate.
3. The oceanic plate is denser as we have already seen, than the continental plate.
4. The oceanic plate is forced underneath the continental plate.
5. The point, at which this happens is called the Subduction Zone.
6. As the oceanic plate is forced below the continental plate, it melts to form magma and earthquakes are triggered.
7. The magma collects to form a magma chamber.
8. The magma then rises up through the cracks in the continental crust.
9. When the pressure builds up, a volcanic eruption may occur.

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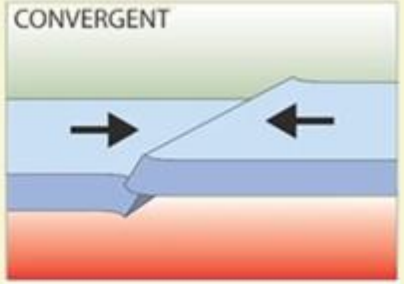
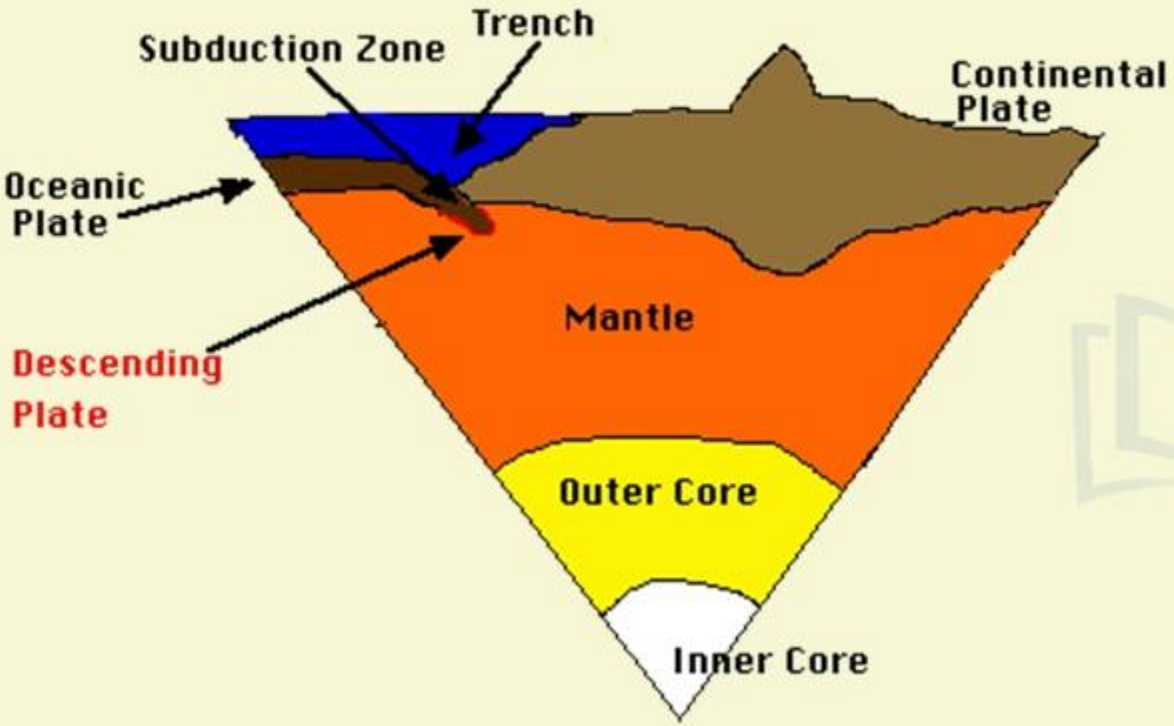
PLATE TECTONICS

2 DESTRUCTIVE / CONVERGENT BOUNDARY

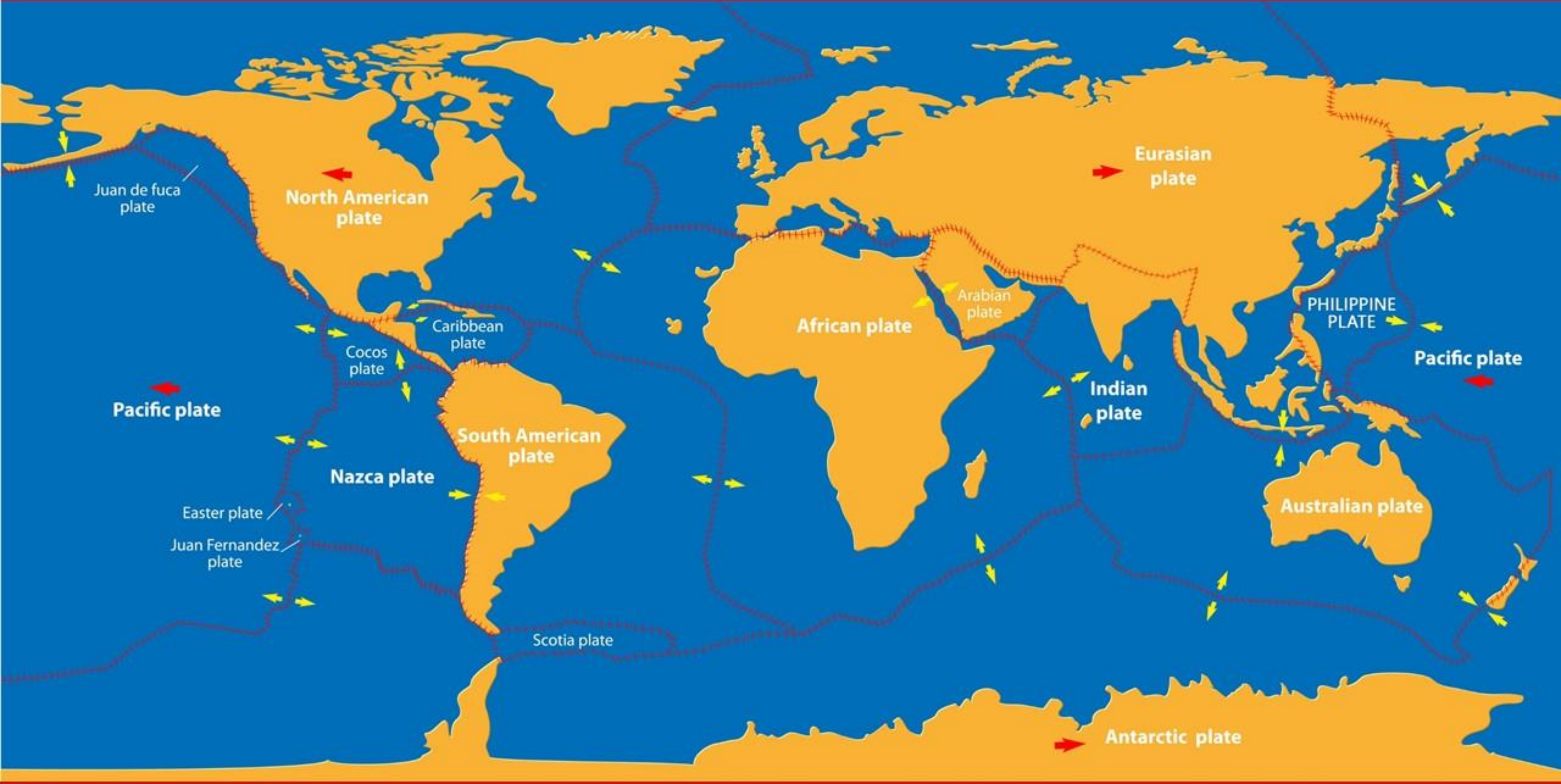


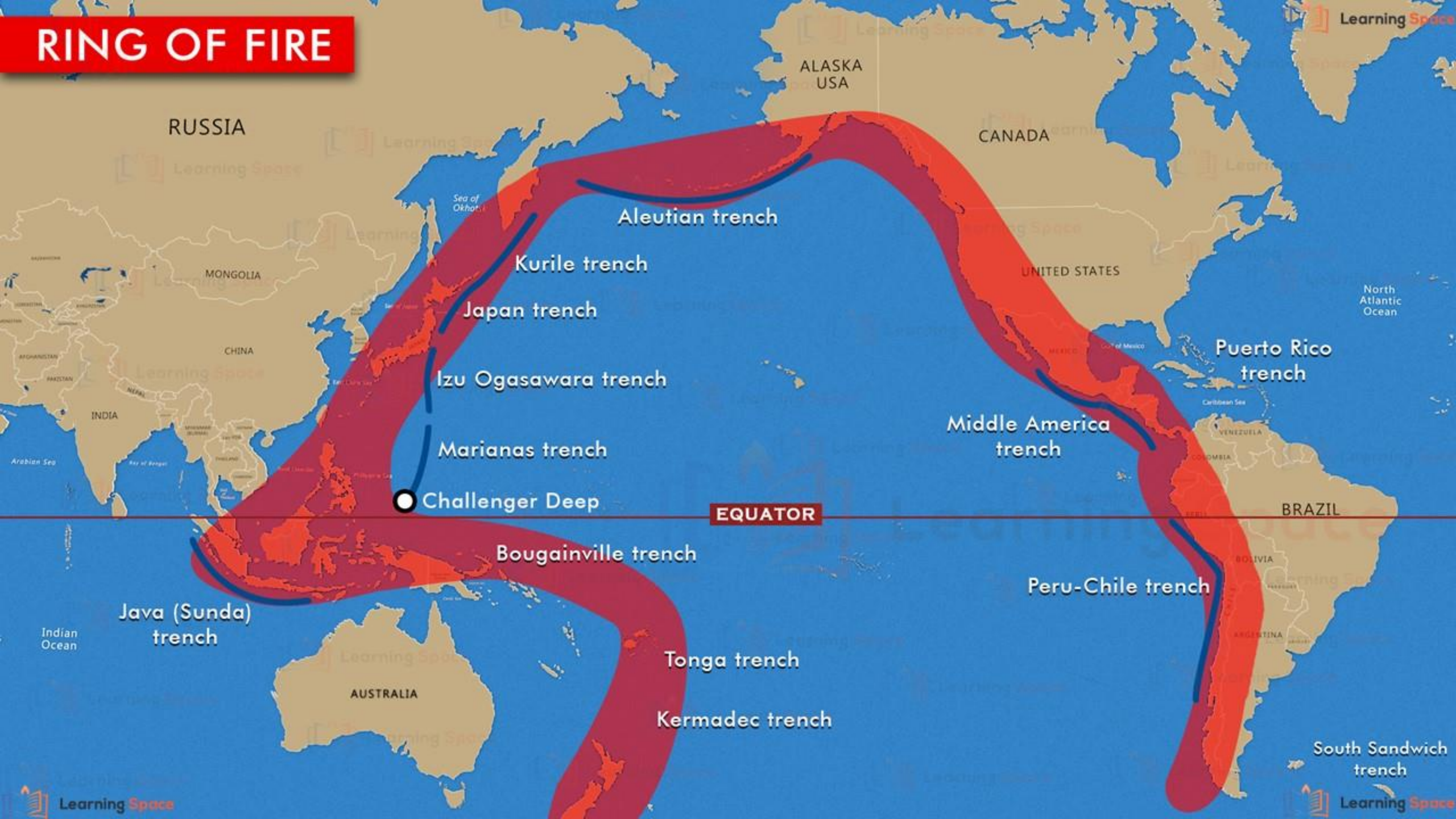
SUBDUCTION ZONES



1. Subduction Zones are the boundaries that mark the collision between two of the planet's Tectonic Plates.
2. We have already seen Tectonic Plates are moving slowly across the planet's surface over millions of years.
3. When two Tectonic Plates meet at a subduction zone, one bends and slides underneath the other, so they curve down into the mantle. Mantle is the hotter layer under the crust.

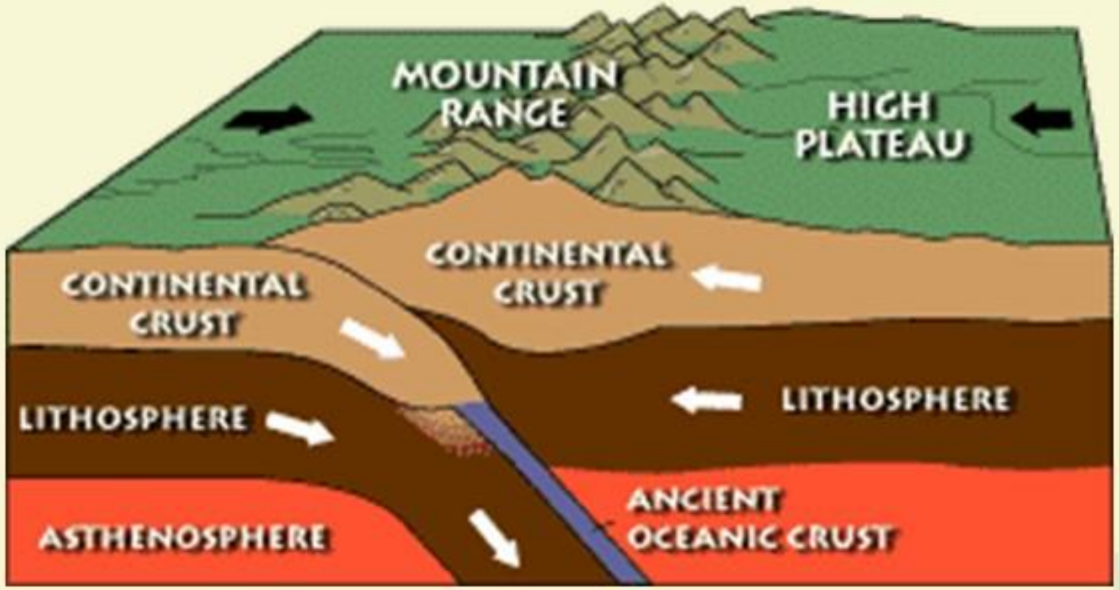
TECTONIC PLATES





RING OF FIRE
RUSSIA
ALASKA
USA
CANADA
MONGOLIA
CHINA
INDIA
UNITED STATES
North
Atlantic
Ocean
Aleutian trench
Kurile trench
Japan trench
Izu Ogasawara trench
Marianas trench
Challenger Deep
EQUATOR
Bougainville trench
Java (Sunda)
trench
Indian
Ocean
Tonga trench
Kermadec trench
AUSTRALIA
Puerto Rico
trench
Middle America
trench
BRAZIL
Peru-Chile trench
South Sandwich
trench

WHAT HAPPENS WHEN A CONTINENT-CONTINENT PLATE COLLIDES?



1. When Continent-Continent plate collides, they may crash together without subducting and crumple together like the crashing cars.
2. The classic example is Himalayas as they were formed, when Indian Plate slammed into Asian Plate.

ASIAN PLATE
INDIAN PLATE
MOUNTAIN RANGES OF THE HIMALAYAS
NEPAL
CHINA
IRAN
AFGHANISTAN
PAKISTAN
UZBEKISTAN
KYRGYZSTAN
TURKMENISTAN
TAJIIKISTAN
AZERBAIJAN
Caspian Sea
Persian Gulf
Gulf of Oman
ARABIA
OMAN
YEMEN
Arabian Sea
Bay of Bengal
BANGLADESH
MYANMAR (BURMA)
VIETNAM
Lao PDR
THAILAND
CAMBODIA
Gulf of Thailand
South China Sea
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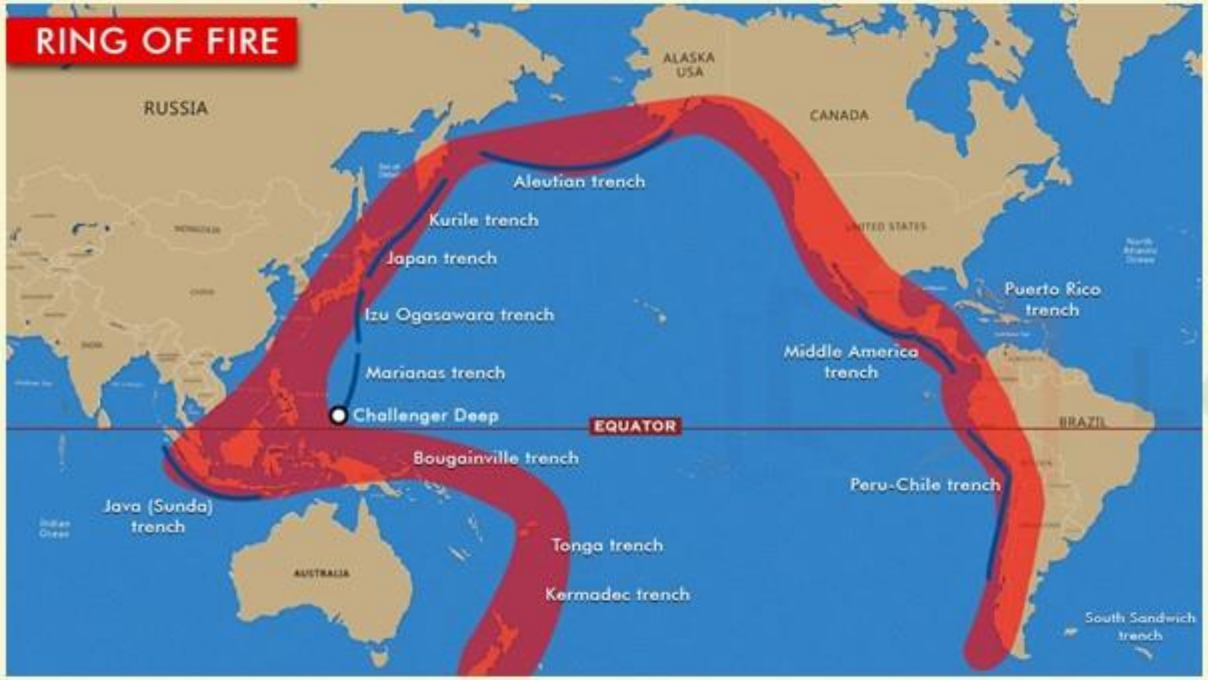
WHERE ARE THE PROMINENT PLACES OF SUBDUCTION ZONES?



- Subduction zones occur all around the edge of the Pacific Ocean. Some of the offshore areas are in Chile, Peru, USA, Canada, Russia, Japan and Indonesia.
- It is called the Ring of Fire.
- These subduction zones are responsible for the world's biggest earthquakes, terrible tsunamis and some of the worst volcanic eruptions.

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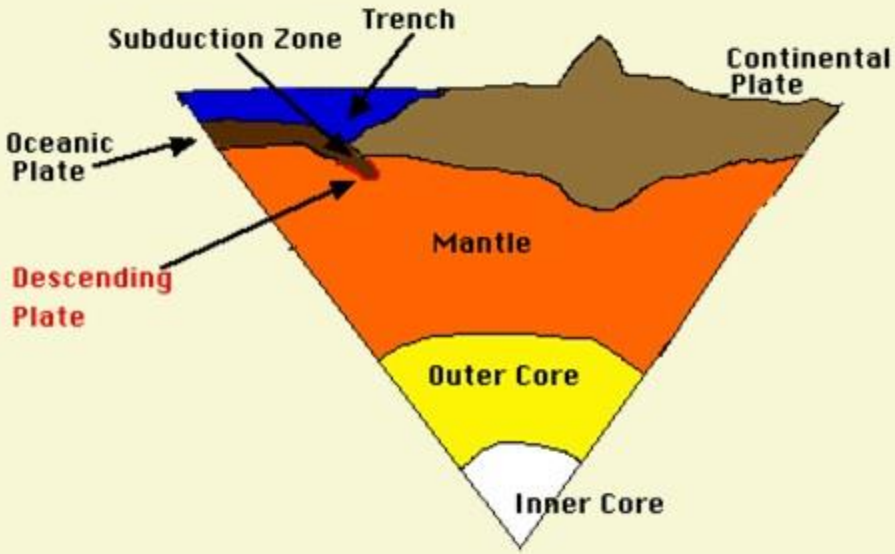
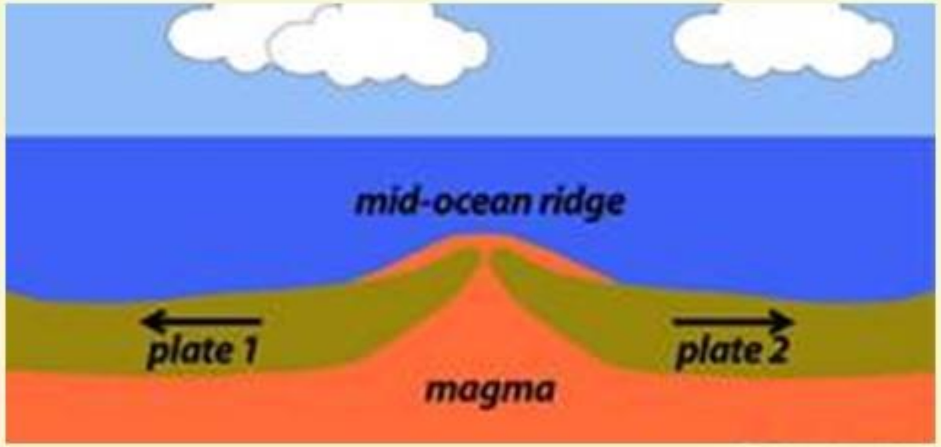
IS THE SUBDUCTION ZONE OPPOSITE TO MID OCEANIC RIDGES?





1. In Subduction Zones, the Oceanic Plate goes down into the Earth's mantle. Hence it is Destructive / Convergent Boundary. Recent studies indicate that the subducting plate is pulled downward by gravity also.

2. However, in the mid oceanic ridges, seafloor is generated from the upper mantle. It spreads laterally outward. It is Divergent / Constructive Boundary.

3. Hence, we can say that the Mid-Ocean Ridges and Subduction Zones are quite opposite phenomenas.

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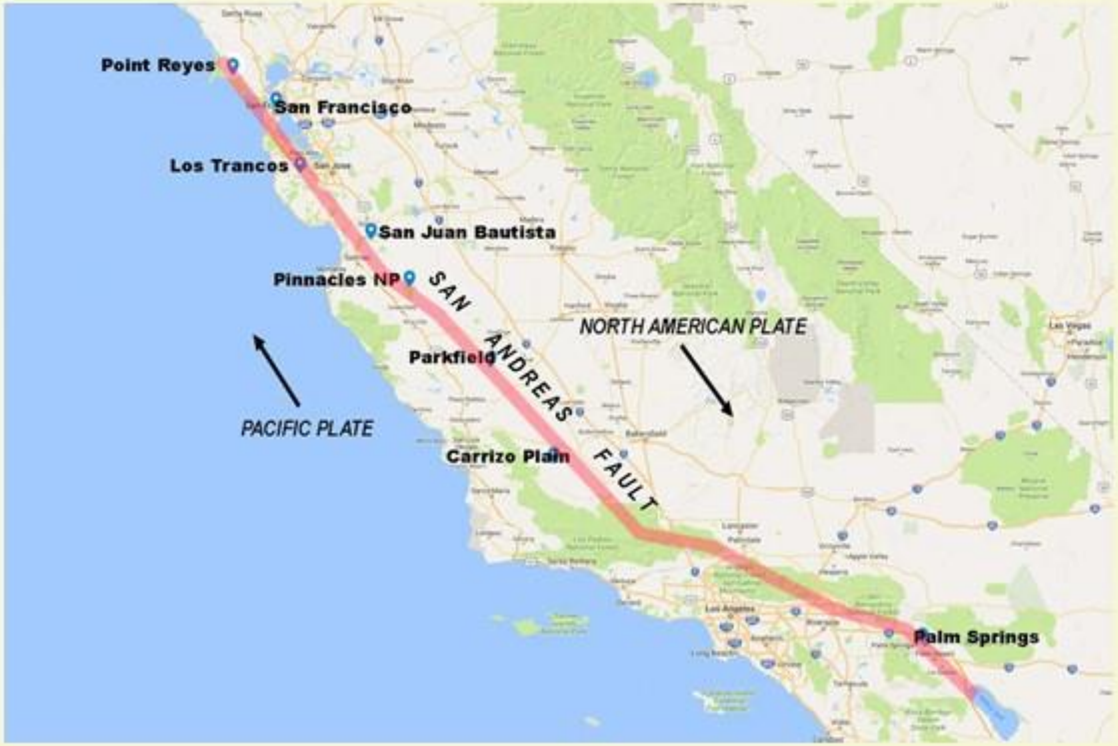
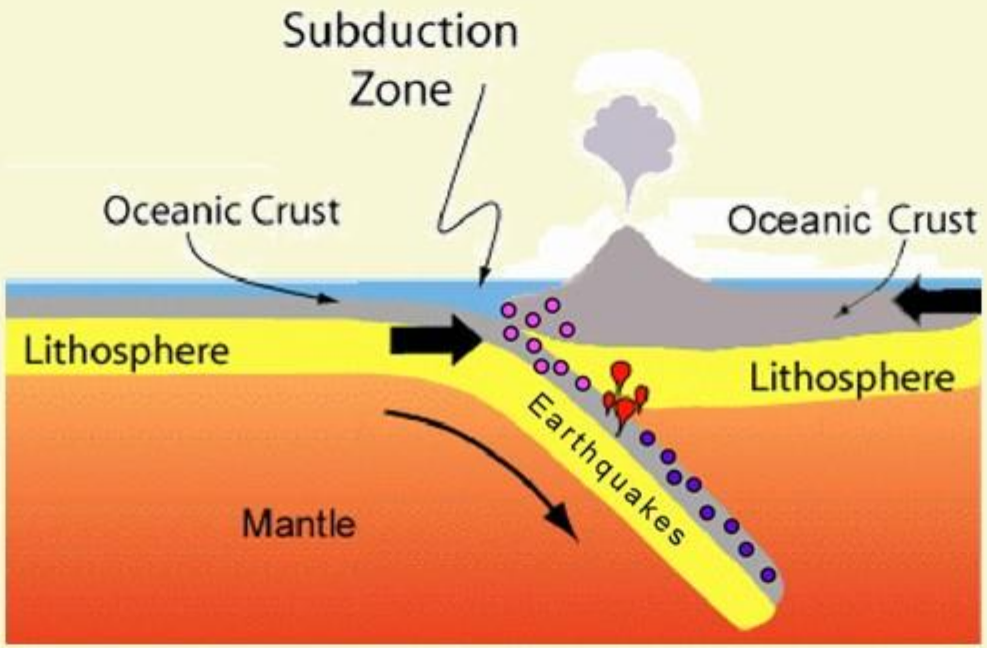
WHAT DO YOU KNOW ABOUT ALASKA SUBDUCTION ZONE?

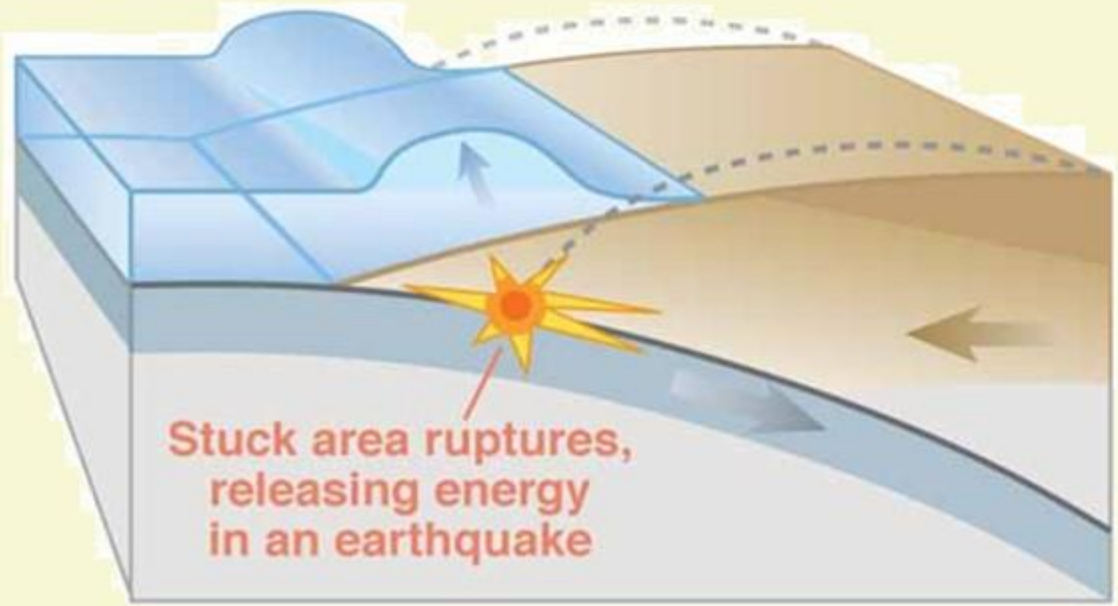


1. It stretches from Russia in the East to Alaska in the West.
2. The Pacific Plate and the North American Plate are moving towards one another at the rate of 6 to 7 Cm / year.
3. The Pacific Plate is thinner and denser, so it is being thrust underneath the North American Plate.
4. This Subduction Zone has generated many large devastating earthquakes, including the second largest earthquake ever recorded at a magnitude of 9.2 and was recorded in 1964 and it caused devastating damage in Alaskan towns and deaths occurred in as far away as California also.


Arctic Ocean
East Siberian Sea
Chukchi Sea
RUSSIA
ALASKA
USA
CANADA
Bering Sea
Sea of Okhotsk
North Pacific Ocean

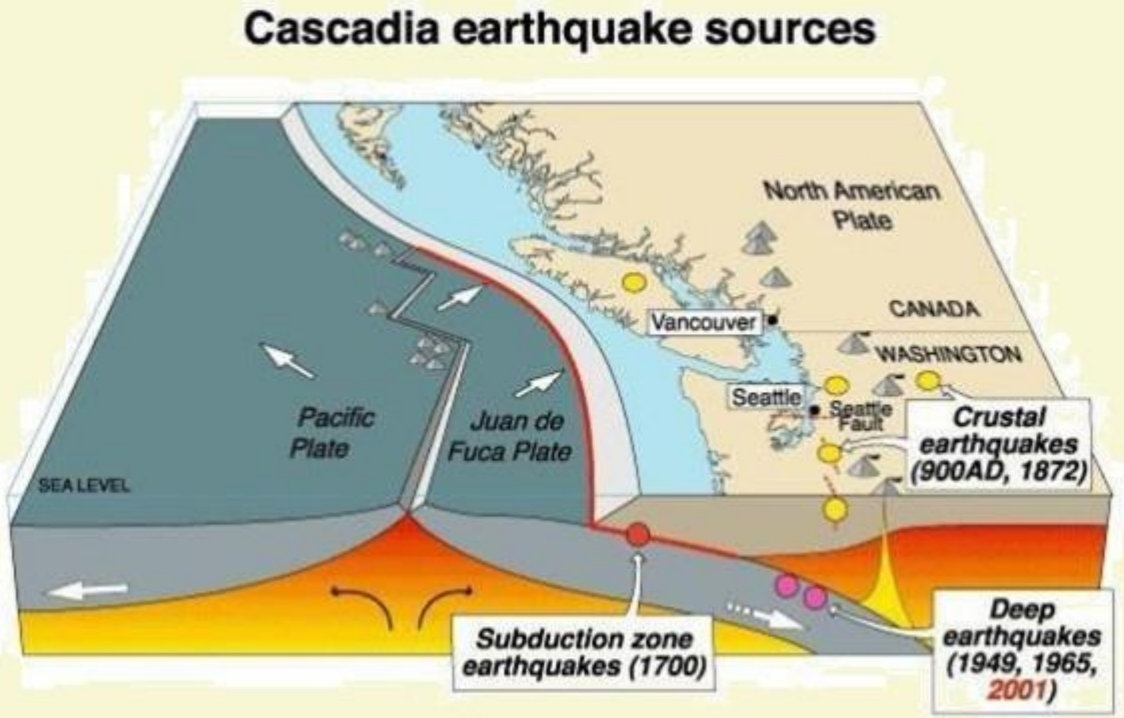
1964 EARTHQUAKE IN ALASKA





SUBDUCTION ZONES AND EARTHQUAKES

1. It is just like rubbing two pieces of sand paper against each other.
2. The crust sticks in some places and it stores up energy and releases it in the form of earthquakes.
3. The size of an earthquake is related to the size of the fault.



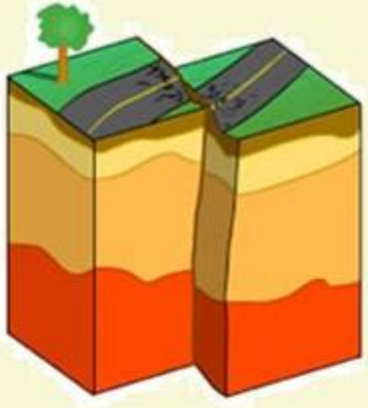
SUBDUCTION ZONES AND EARTHQUAKES

4 The subduction zone faults are the longest and widest in the world.

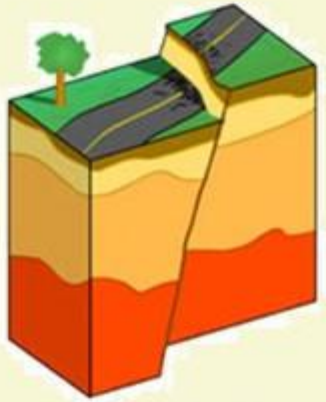
5 The Cascadia Subduction Zone of shore of Washington is about 1,000 Kms long and 100 Kms wide.

6 Smaller earthquakes also strike all along the descending plate, which is also called a slab.

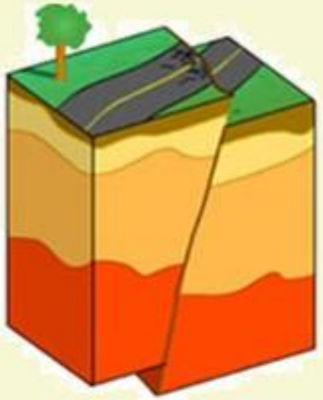
WHAT DO YOU UNDERSTAND BY FAULTS?



STRIKE-SLIP FAULT



NORMAL FAULT



THRUST / REVERSE FAULT

1. Faults are fractures in Earth's crust, where the rocks on either side of the crack have slid past each other.

2. Sometimes, the cracks are tiny with barely noticeable movement between the rock layers.

3. But, the faults can be hundreds of kilometres long, such as San Andreas Fault in California and the Anatolian Fault in Turkey.

4. The faults are three types. They are ---
   - Strike slip faults
   - Normal faults
   - Thrust faults

   Each type is the outcome of different forces i.e., pushing or pulling on the crust causing the rocks to slide up or down or pass each other.

SAN ANDREAS FAULT, CALIFORNIA

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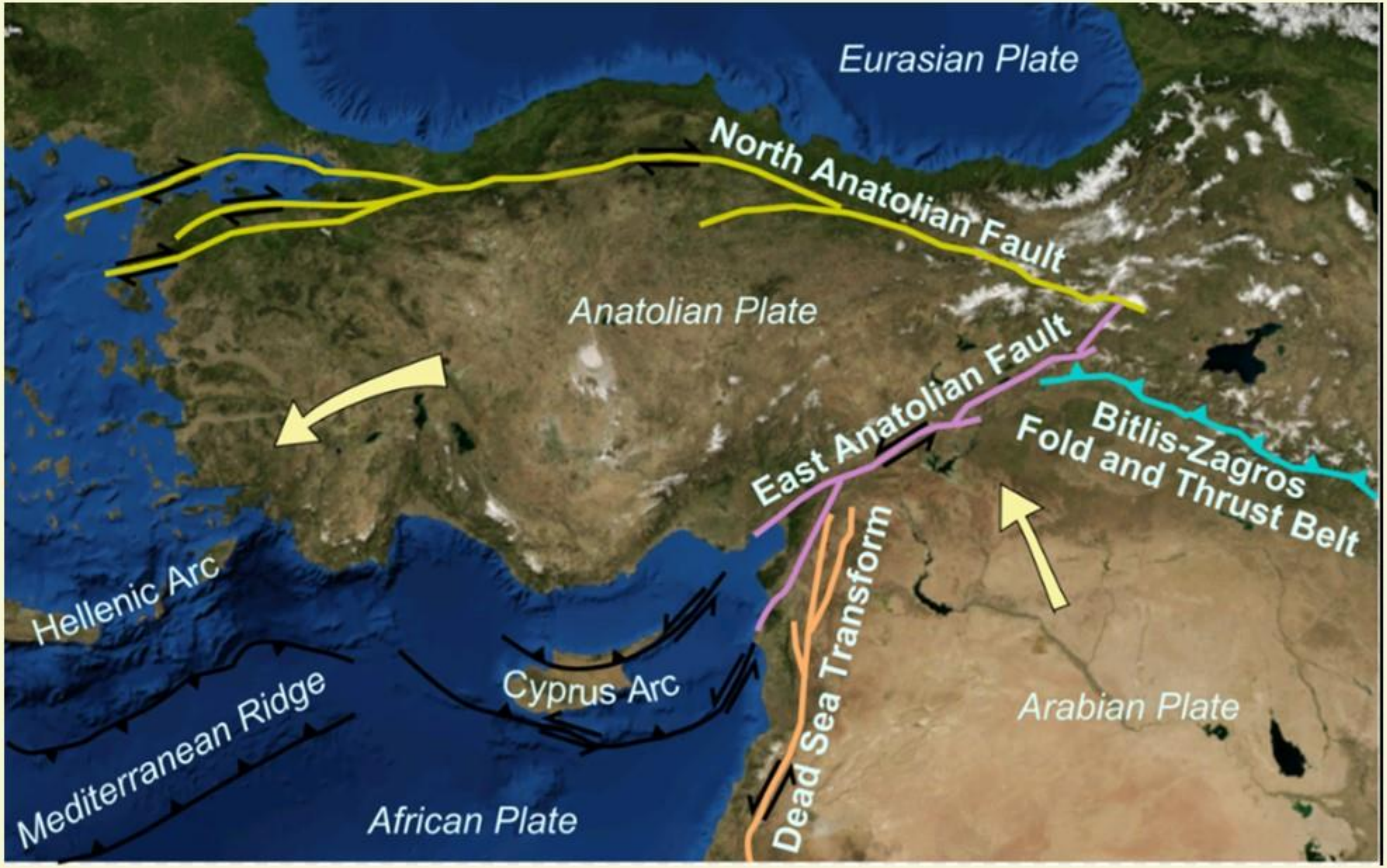
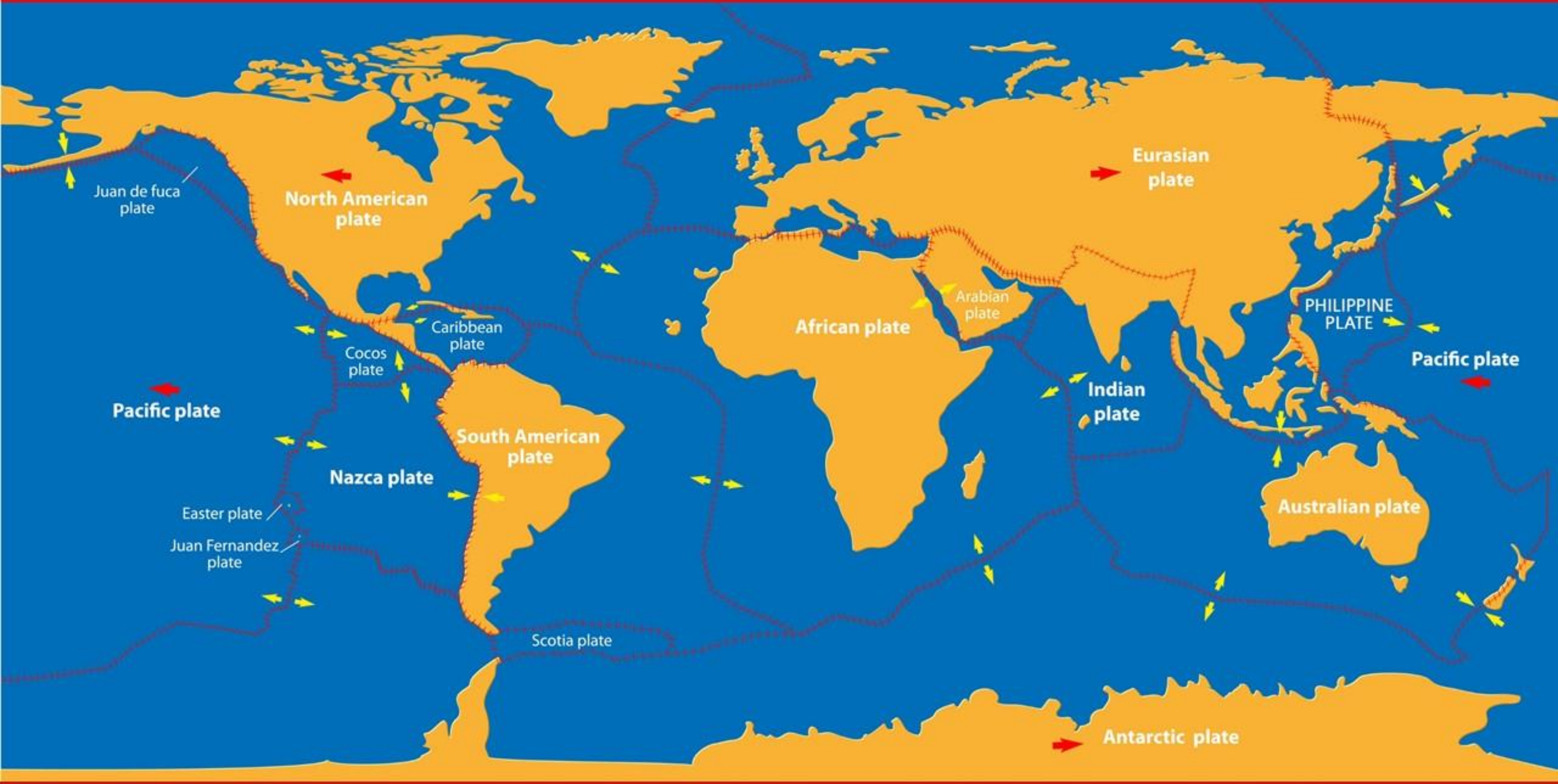
ANATOLIAN FAULT, TURKEY



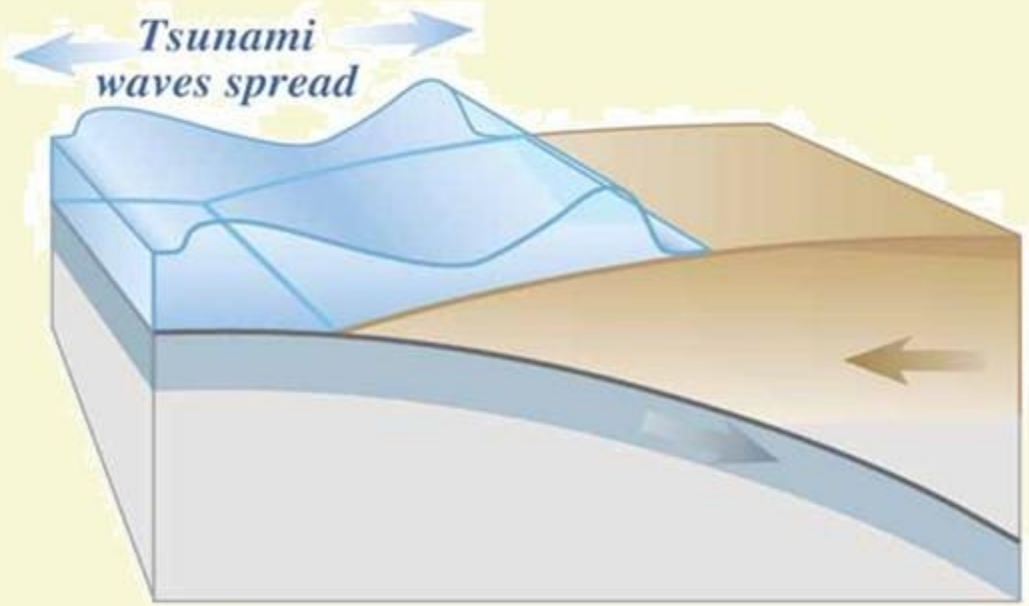
MORE ON THE FAULTS

1. Faults are related to the movement of Earth's Tectonic Plates.
2. The biggest faults mark the boundary between the two plates.
3. The boundary between the two plates are broad zones of deformation.
4. Plate boundaries are always growing and changing, so these faults develop kinks and bends as they slide past each other, which generates more faults.

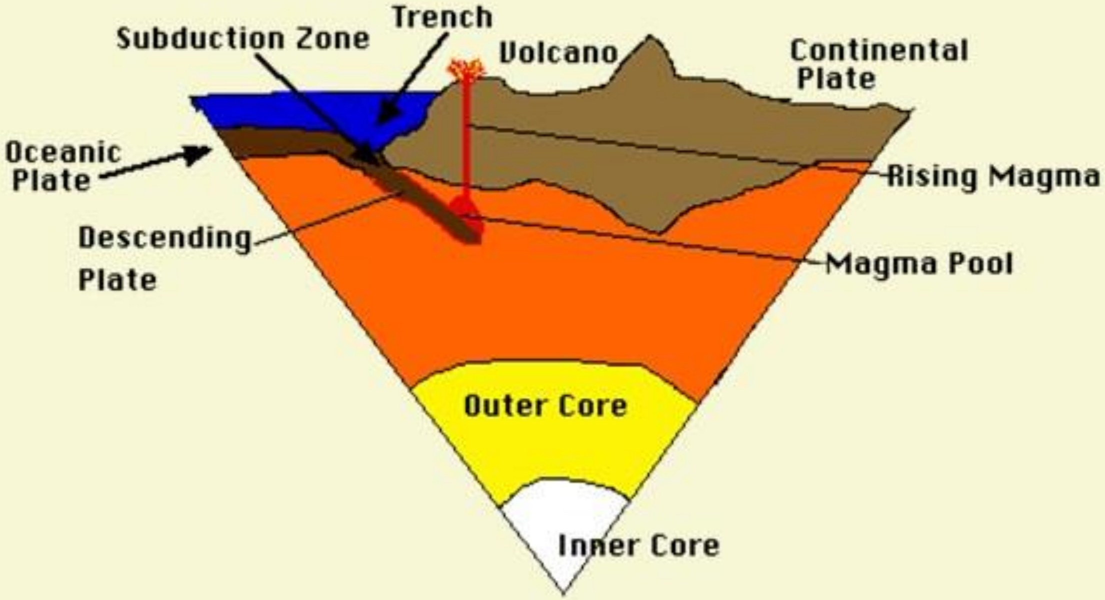
TECTONIC PLATES



WHY TSUNAMIS ARE ALSO CAUSED?

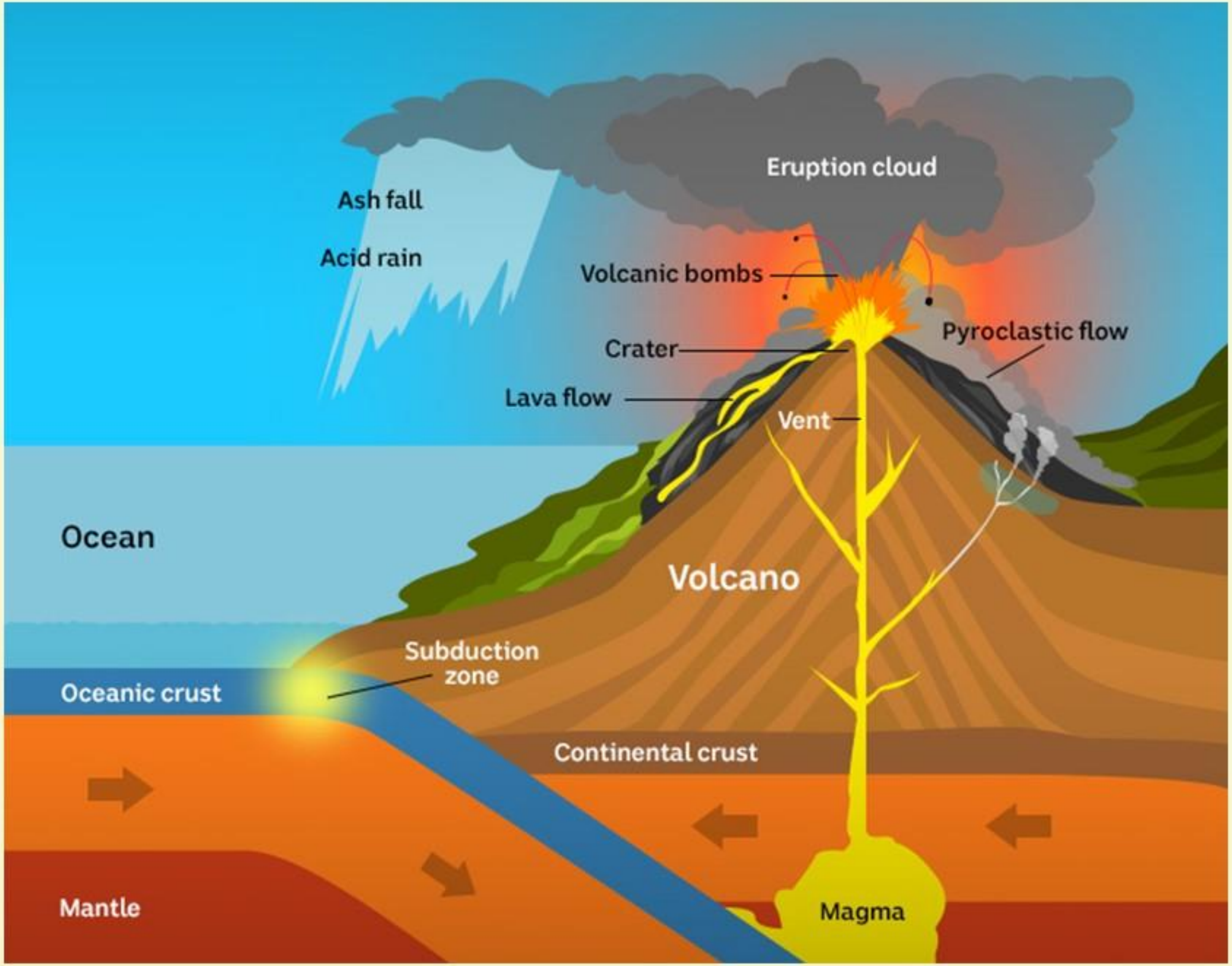


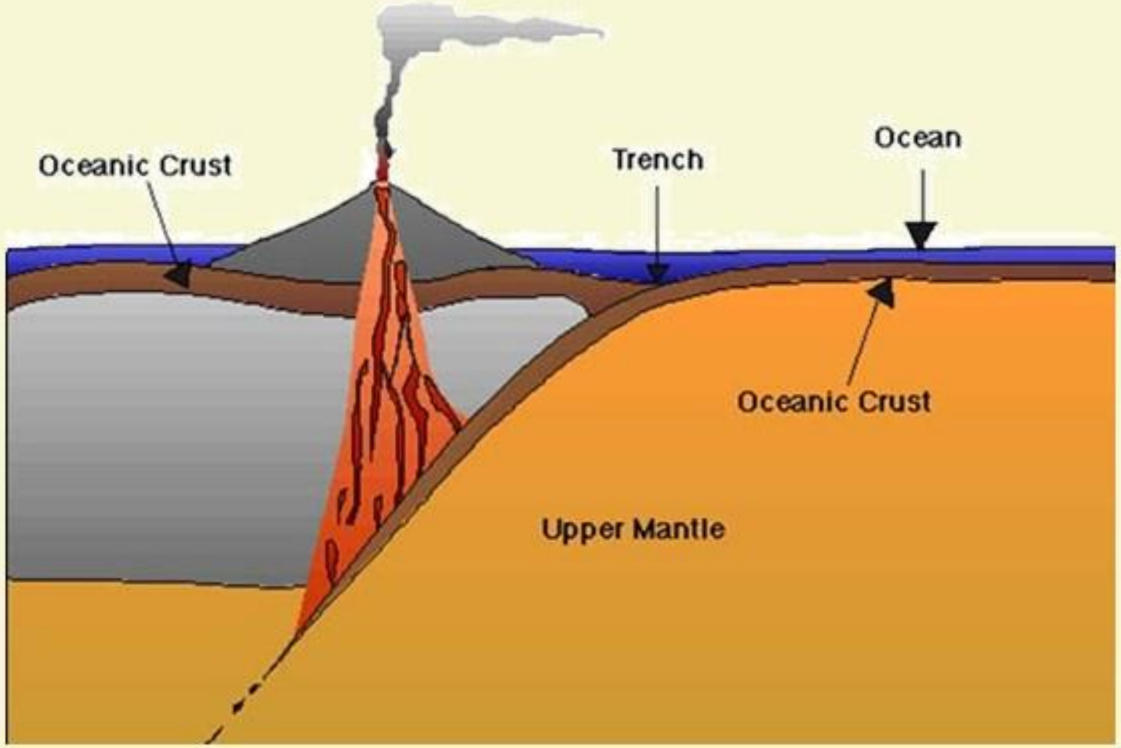
1. Most of the subduction zones are usually along the coastlines.
2. So Tsunamis will always be generated close to where people live.
3. When subduction zone earthquakes hit, Earth's crust flexes and snaps like a freed spring.
4. For the magnitude of 7.5 and above, it can cause a Tsunami by suddenly moving the seafloor. It is nothing but, a giant sea wave.
5. Not all subduction zone earthquakes will cause tsunamis. Some earthquakes also trigger tsunamis by sparking underwater landslides.



WHY VOLCANOES OCCUR AT THE SUBDUCTION ZONES?

1. As the Tectonic Plate slides into the mantle, the hotter layer beneath Earth's crust, the heating releases fluids trapped in the plate.
2. These fluids, such as seawater and carbon dioxide, rise into the upper plate and can partially melt the overlying crust, forming magma and magma (molten rock) often means volcanoes.


Eruption cloud
Ash fall
Acid rain
Volcanic bombs
Pyroclastic flow
Crater
Lava flow
Vent
Ocean
Volcano
Subduction zone
Oceanic crust
Continental crust
Mantle
Magma



WHAT ARE THE EXAMPLES?

1. There is a link between subduction zones and volcanoes, when one looks at several places across the globe.
2. Inland of each subduction zone is a chain of spouting volcanoes called a volcanic arc, such as Alaska's Aleutian Islands.
3. The Toba volcanic eruption in Indonesia is one of the largest volcanic eruptions. It is also from a subduction zone.


Arctic Ocean
East
Siberian Sea
Chukchi Sea
RUSSIA
ALASKA
U.S.A
CANADA
Bering Sea
Sea of
Okhotsk
ALEUTIAN ISLANDS
North Pacific Ocean


INDIA
MYANMAR (BURMA)
VIETNAM
Lao PDR
THAILAND
CAMBODIA
Bay of Bengal
Andaman Sea
Gulf of Thailand
South China Sea
MALAYSIA
Lake Toba
SUMATRA

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